

10/20/2025 Briana-Yves 1-1 Meeting Minutes

We reviewed the Deep Dream results. To improve the model's interpretation of tissue texture, I suggested fine-tuning the algorithm with alternative loss functions, such as a tumor-focused top-k activation loss (top 5% of most active neurons) or a segmentation-based loss to better highlight tumor regions.

Next steps:

- Run Deep Dream on different encoder or decoder layers (early layers emphasize fine texture; deeper layers capture semantic tumor structures).
- Combine activations from multiple layers to balance texture detail and regional context.
- Add spatial smoothness or total-variation penalties to reduce noise and make textures more biologically interpretable.

I tried a Grad-CAM approach; Layer-CAM with less success.

